

Pinhole Camera

By Neeti patel of class 6-H

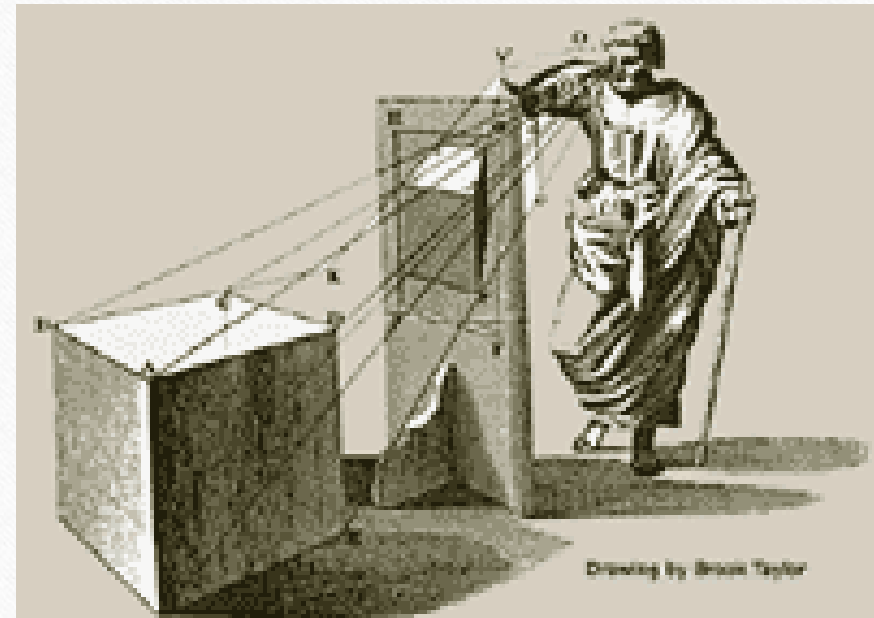
What is Pinhole Camera?

- A simple optical device that uses a tiny hole to project an inverted image on a surface.
- No lens required; works on the principle of light traveling in straight lines.
- Oldest known form of a camera, dating back to ancient times.



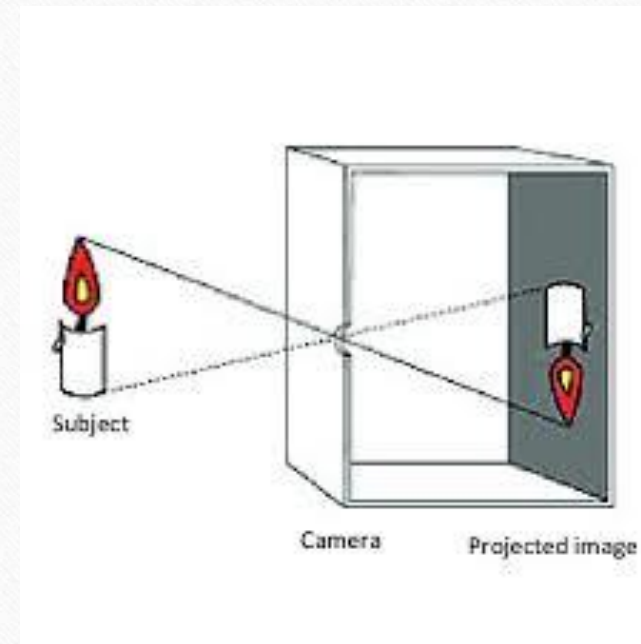
History of the Pinhole Camera

- Origin: Known as the "camera obscura," Latin for "dark chamber."
- Used by ancient Greek and Chinese philosophers for observing solar eclipses.
- Revolutionized by Renaissance artists for accurate perspective drawings.



Principle of Pinhole Camera

- Relies on the principle of rectilinear propagation of light.
- Steps:
 - Light enters through a small hole.
 - An inverted image forms on the opposite side of the box or surface.
- Key Features:
 - No lens or complex machinery.
 - Works best in bright lighting conditions.



Building a Pinhole Camera

- Materials Needed:
- A cardboard box or opaque container.
- Aluminum foil or thin metal sheet.
- Black paper to block external light.
- Wax paper or tracing paper as the screen.
- Tape, scissors, and a needle for the pinhole.



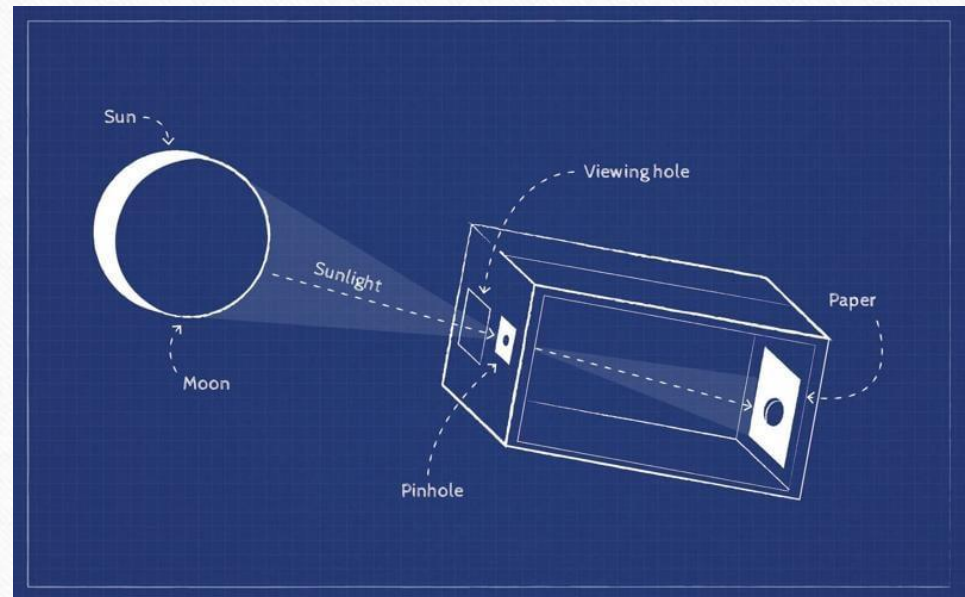
Uses of a Pinhole Camera

- Educational tool to demonstrate basic optics principles.
- Safe observation of solar eclipses.
- Photography: Captures unique, soft-focus images.
- Artistic experiments in light and shadow.



Pros and cons of the Pinhole Camera

- **Advantages:**
- Simple and cost-effective.
- Does not require a lens.
- Offers infinite depth of field.
- **Limitations:**
- Low brightness of the image.
- Requires long exposure time for photography.
- Limited resolution compared to modern cameras.



Thank You 😊